

ELEC 221 Lecture 07

CT Fourier series and filters

Thursday 1 October 2026

Announcements

- Tutorial Assignment 2 on Monday: bring a short (3-5) music sample in .wav format
- Quiz 4 on Tuesday

Last time

Fourier synthesis equation:

Fourier analysis equation:

Last time

We evaluated the Fourier series coefficients of

$$x(t) = e^{-t}, \quad -1 \leq t < 1$$

We got an exact result

$$c_k = \frac{(-1)^k}{2(1 + jk\pi)} [e - e^{-1}]$$

Last time

Can we *always* express a signal as a Fourier series?

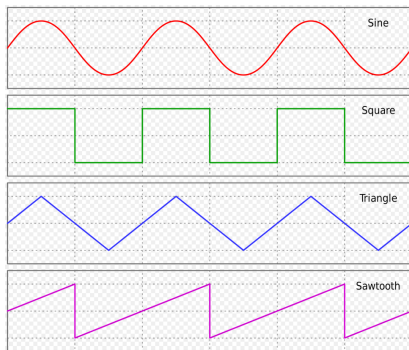


Image credit: *Sine, square, triangle, and sawtooth waveforms* (author: Omegatron)

https://en.wikipedia.org/wiki/Triangle_wave#/media/File:Waveforms.svg (CC BY-SA 3.0)

Last time

Given a **periodic function**, if over one period,

- ① is single-valued
- ② is absolutely integrable
- ③ has a finite number of maxima and minima
- ④ has a finite number of discontinuities

then the Fourier series converges to

- $x(t)$ where it is continuous
- half the value of the jump where it is discontinuous

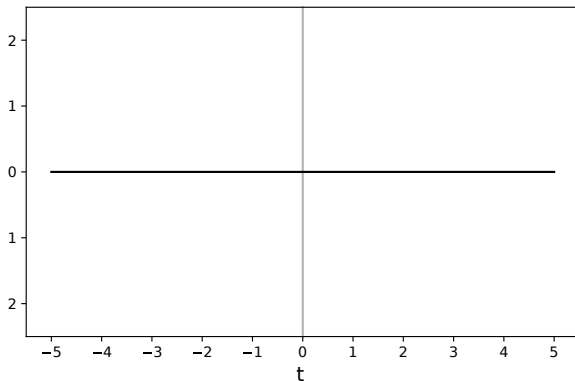
Today

Learning outcomes:

- Evaluate the Fourier series coefficient of a square wave; describe the Gibbs phenomenon
- State and apply key properties of CT Fourier series
- Express filters using the frequency response

Example: the square wave

$$x(t) = \begin{cases} 1, & 0 < t < \pi, \\ -1, & \pi < t < 2\pi \end{cases}$$



Example: the square wave

Evaluate its Fourier coefficients. We can often take shortcuts based on the properties of a signal.

You can derive that for a 2π -periodic function, the coefficients have the following form:

Example: the square wave

Since the function is symmetric, we can rewrite this:

Will it converge?

Convergence: signal power and energy

Define the **total energy** of a signal over the period t_1 to t_2 as

The time-averaged **power** is

Signals with finite energy over a period can be represented using a Fourier series.

Convergence: signal power and energy

First, derive an important relationship between our signal and its Fourier coefficients (**Parseval's relation**):

Convergence: signal power and energy

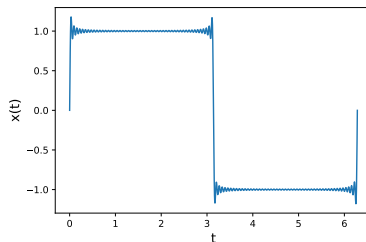
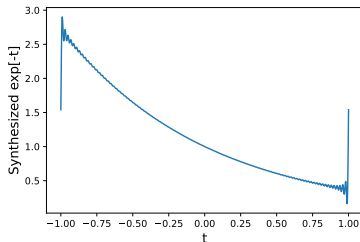
Imagine truncating the sum at N :

Consider approximation error of the truncation:

How much energy in error is there over a given period?

Take limit as $N \rightarrow \infty \dots$

Gibb's phenomenon



The amount of ringing, or “overshoot”, is about 9% of the jump of the discontinuity, no matter where we truncate.

Watch a derivation here:

<https://www.youtube.com/watch?v=MvXY8MXSit4>

Properties of Fourier series

In the square wave example, we leveraged some shortcuts to compute the Fourier coefficients.

They have other useful properties that help evaluate them.

Let's see what happens to the Fourier series when we apply:

- Superposition
- Time shift / scale / reversal
- Multiplication

Properties of Fourier series

Fourier coefficients combine linearly.

Suppose we have two signals $x(t), y(t)$ with period T ,

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t},$$
$$y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk\omega t},$$

Then the signal $z(t) = Ax(t) + By(t)$ has the form

Properties of Fourier series

Time shift $x(t) \rightarrow x(t - t_0)$:

Properties of Fourier series

Time scale $x(t) \rightarrow x(\alpha t)$.

If original period was T , new period is T/α :

Properties of Fourier series

Multiplication leads to convolution of the coefficients:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t},$$

$$y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk\omega t},$$

Then the signal $z(t) = x(t)y(t)$ has the form

Properties of Fourier series

TABLE 3.1 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

Property	Section	Periodic Signal	Fourier Series Coefficients
		$\left. \begin{array}{l} x(t) \\ y(t) \end{array} \right\} \begin{array}{l} \text{Periodic with period } T \text{ and} \\ \text{fundamental frequency } \omega_0 = 2\pi/T \end{array}$	$\begin{array}{l} a_k \\ b_k \end{array}$
Linearity	3.5.1	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	3.5.2	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting		$e^{jM\omega_0 t} = e^{jM(2\pi/T)t} x(t)$	a_{k-M}
Conjugation	3.5.6	$x^*(t)$	a_{-k}^*
Time Reversal	3.5.3	$x(-t)$	a_{-k}
Time Scaling	3.5.4	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution		$\int_T x(\tau)y(t - \tau)d\tau$	$T a_k b_k$
Multiplication	3.5.5	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$
Differentiation		$\frac{dx(t)}{dt}$	$jk\omega_0 a_k = jk \frac{2\pi}{T} a_k$
Integration		$\int_{-\infty}^t x(t) dt$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0}\right)a_k = \left(\frac{1}{jk(2\pi/T)}\right)a_k$
Conjugate Symmetry for Real Signals	3.5.6	$x(t)$ real	$\begin{cases} a_k = a_{-k}^* \\ \Re\{a_k\} = \Re\{a_{-k}\} \\ \Im\{a_k\} = -\Im\{a_{-k}\} \\ a_k = a_{-k} \\ \angle a_k = -\angle a_{-k} \end{cases}$
Real and Even Signals	3.5.6	$x(t)$ real and even	a_k real and even
Real and Odd Signals	3.5.6	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals		$\begin{cases} x_e(t) = \mathcal{E}\{x(t)\} & [x(t) \text{ real}] \\ x_o(t) = \mathcal{O}\{x(t)\} & [x(t) \text{ real}] \end{cases}$	$\begin{cases} \Re\{a_k\} \\ j\Im\{a_k\} \end{cases}$

Exercise

Go back to the square wave

$$x(t) = \begin{cases} 1, & 0 < t < \pi, \\ -1, & \pi < t < 2\pi \end{cases}$$

We obtained

$$x(t) = \sum_{k=1}^{\infty} b_k \sin(kt), \quad b_k = \begin{cases} 0, & k \text{ is even} \\ 4/k\pi, & k \text{ is odd} \end{cases}$$

What are the Fourier coefficients of the square wave

$$x(t) = \begin{cases} 1, & -\frac{\pi}{4} < t < \frac{\pi}{4}, \\ -1, & \frac{\pi}{4} < t < \frac{3\pi}{4} \end{cases}$$

Exercise

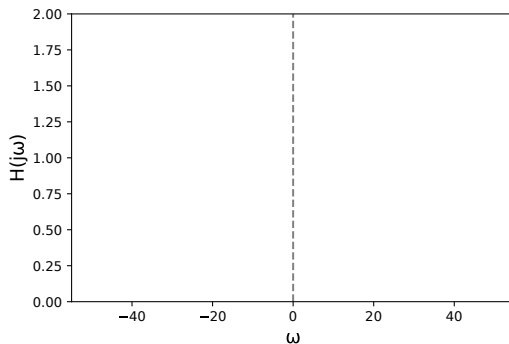
Step 1: express the b_k as the “original” coefficients c_k

Exercise

Step 2: apply the transformations

Example

What does a system with the following frequency response do?



Filters

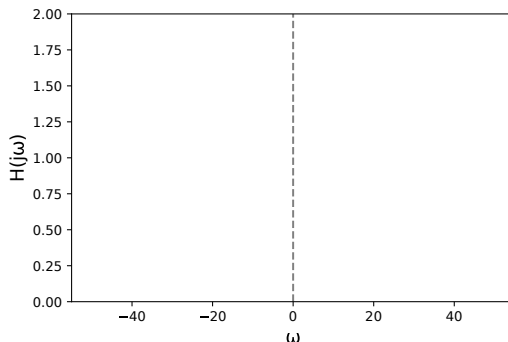
Filters are LTI systems that can be used to separate out, combine, or modify the components of a signal at specific frequencies.

Two key types:

- **Frequency-shaping**: change the amplitudes of parts of a signal at specified frequencies
- **Frequency-selective**: eliminate or attenuate parts of a signal at specified frequencies

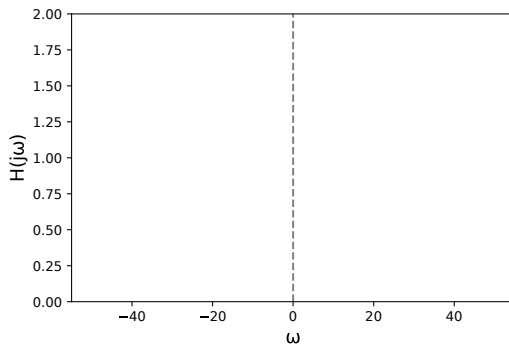
CT frequency-selective filters

We can also consider an ideal **highpass filter**:



CT frequency-selective filters

Or an ideal **bandpass** filter:



Lowpass filters in practice

Some filters made of physical components are described by differential equations.

Example: a lowpass filter using a resistor and a capacitor.

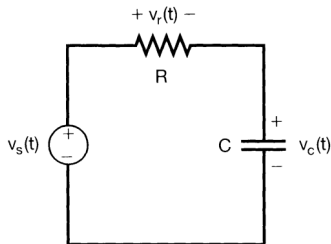
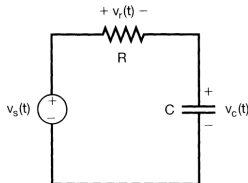


Image credit: Oppenheim chapter 3.10.

Exercise: lowpass filters in practice

What is the voltage across the capacitor if $v_s = e^{j\omega t}$?



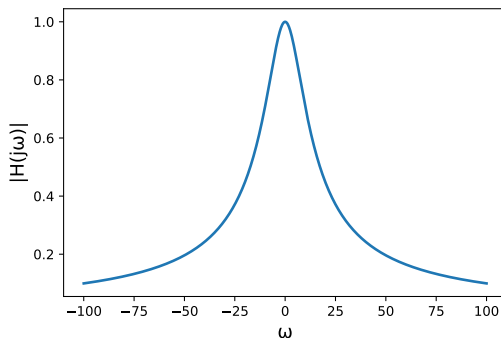
Can derive two expressions for the current, using the resistor and capacitor respectively:

Exercise: lowpass filters in practice

Put these together to form a differential equation:

Exercise: lowpass filters in practice

Results in the following frequency response (setting $RC = 0.1$):



Adjusting the value of RC controls the frequency response. Increasing RC cuts off more frequencies.

For next time

Content:

- DT periodic signals
- DT Fourier series coefficients
- DT frequency response and filters

Action items:

- ① Work on practice assignments 1 and 2
- ② Choose music for tutorial assignment

Recommended reading:

- From today's class: Oppenheim 3.3-3.5, 3.9
- Suggested problems: 3.1, 3.3-3.8, 3.13, 3.15, 3.19, 3.21, 3.22a,c, 3.23-3.26, 3.33, 3.35, 3.43-3.46
- For next class: Oppenheim 3.6-3.7